

Validity Analysis: LTZGLUE

Target context: Luxembourgish Professional Newsroom

Overall risk: **HIGH**

Deployment Context

Use case: In a newsroom setting, journalists and editors use LLM-powered software to manage Luxembourgish news articles by applying a model evaluated on headline acceptability, linguistic acceptability, topic classification, and named entity recognition. The system automatically verifies grammatical standards, classifies the article's domain, and identifies key entities to enable automatic internal linking between related news stories. These outputs determine whether a headline is logically consistent with its content and if the article can be instantly published or must be flagged for manual editorial revision to maintain professional standards.

Target population: Professional journalists, editors, and digital content managers at Luxembourgish media outlets who produce and curate news content in the Luxembourgish language for a local audience.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content △	2	Significant gaps	medium
Input Form	3	Moderate gaps	medium
Output Ontology △	2	Significant gaps	high
Output Content △	2	Significant gaps	high
Output Form ✓	4	Minor gaps	high
Average	2.7		

△ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

LTZGLUE is the first GLUE-style benchmark for Luxembourgish and covers all four tasks the newsroom deployment requires (HA, LA, TC, NER), with RTL — the deployment's primary outlet — as principal data source. However, three of the highest-priority deployment dimensions (IC code-switching, OO output ontology, OC annotator population) show substantial

misalignment. Code-switching, a HIGH-priority operational requirement, is acknowledged as under-represented [Q112] and only the NER comment subcorpus is documented as covering it [Q41]. The LA Ortho ground truth is derived from Spellchecker.lu encoding the formal ZLS standard [Q31, WEB-11], creating systematic divergence from the deployment's flexible journalistic register [A3]. Annotators are student assistants [Q114] with low documented IAA where measured (Cohen's Kappa 0.45 for SA [Q27]) and AI-assisted labelling for RTE [Q63, Q64] and JUDGEWEL [Q37] — none professional journalists. Output form (OF) is well aligned and partially mitigates these concerns. The benchmark itself explicitly cautions against treating its scores as evidence of cultural or demographic coverage [Q120, Q121].

ID	Evidence
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)
Q37	"The resulting sentences are then evaluated using LLMs acting as judges, with minimal human verification to calibrate quality thresholds." (p.4)
Q41	"It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE)." (p.4)
Q63	"To achieve (a), we prompted CHATGPT-5-MINI to judge the texts in the improved data and label their quality as either low, medium, or high, keeping only data rated at least medium, removing nearly 25% of the entire dataset (see Appendix 7.5)." (p.5)
Q64	"For (b), we prompted CHATGPT-5-MINI to verify whether the dataset labels remained correct after the first translation and improvement, outputting true or false for each sentence pair (see Appendix 7.6). Nearly 10% of the labels were false." (p.5)
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)
Q114	"We would like to thank the student assistants for their annotation work." (p.9)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies (https://data.public.lu/en/organizations/zenter-fir-dletzebuerger-sprooch/)

Practical Guidance

What This Benchmark Measures

LTZGLUE provides a credible first-pass evaluation of generic Luxembourgish NLU capability across the deployment's four required tasks, with strongest signal on output-form compatibility (OF), task-coverage breadth (IO, partially), and an authentic RTL-grounded text source for HA, NER, and TC. Strong encoder performance on TC and NER [Q94, Q105] suggests these tasks are reasonably tractable.

Construct Depth

The benchmark probes surface-level NLU constructs (categorical classification, sequence labelling) but does not probe the deployment's actual ground truth — professional editorial judgment under flexible journalistic register with dense code-switching. LA in particular operationalises formal-orthographic correctness via LOD/Spellchecker [Q30, Q31] rather than journalistic acceptability; HA operationalises TF-IDF structural mismatch [Q22] rather than editorial coherence. Output Content (OC) and Output Ontology (OO) gaps mean high benchmark scores would not directly transfer to editorial-quality judgments.

What Else You Need

To complete the assessment, the deployer should: (a) commission re-annotation of held-out HA and LA samples by professional Luxembourgish editors to address OC and OO gaps; (b) audit benchmark instances for code-switching density

and supplement with a code-switched evaluation set to address the HIGH-priority IC gap; (c) re-evaluate or expand the TC taxonomy against the deploying outlet's actual editorial categories (IO); (d) reinstate GPE/LOC granularity for cross-border NER use cases (OO); (e) calibrate per-task model confidence scores against editorial flagging thresholds in a shadow-deployment phase before live auto-publication.

ID	Evidence
Q22	"The resulting negative examples remain topically related but are temporally and structurally mismatched, forcing models to attend to article content rather than surface cues." (p.3)
Q30	"The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset." (p.3)
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)
Q94	"Encoder-based models perform strongly across most settings, particularly on structurally complex and label-sensitive tasks, confirming findings from prior work on multilingual and low-resource NLU (Wu and Dredze, 2019; Conneau et al., 2020)." (p.7)
Q105	"First, MMBERT-base consistently achieves the strongest or near-strongest performance across almost all tasks, combining high mean F1 scores with comparatively low variance, suggesting that broad multilingual pre-training with sufficient LTZ exposure yields stable and transferable representations." (p.8)

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark covers all four tasks the deployment requires (HA, LA, TC, NER) [Q15, Q29, Q46, Q34], and notably introduces headline acceptability as a novel contribution for Luxembourgish [Q69], directly addressing the deployment's headline-vs-body consistency need. The topic taxonomy maps to standard international news categories aligned with the deployment [Q48, A1]. However, three of the eight tasks (SA, ID, RTE) are operationally irrelevant to the newsroom pipeline, and the topic taxonomy includes ANIMALS [Q48] — an unusual inclusion whose newsroom relevance is unjustified — while plausibly missing standard newsroom categories such as POLITICS, HEALTH, and JUSTICE which Luxembourgish media routinely cover (including EU institutions and Grande Région affairs per [WEB-1]). This represents partial alignment: the required task types are present, but taxonomy gaps within TC may cause misclassification in production.

ID	Evidence
Q15	"We formulate headline acceptability (HA) as a binary classification task where the model must decide whether a given headline matches the accompanying article body." (p.3)
Q29	"We introduce a linguistic acceptability dataset consisting of four distinct linguistic subtypes, which can either be used as a binary (LA (BINARY)) or multiclass (LA (MULTI)) classification dataset." (p.3)
Q34	"The JUDGEWEL dataset (Plum et al., 2026) introduces an automatically constructed corpus for named entity recognition (NER) in LTZ, derived from Wikipedia and Wikidata." (p.4)
Q46	"To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories." (p.4)
Q48	"From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS." (p.4)
Q69	"In addition, a substantial proportion of the LTZGLUE tasks are newly created for LTZ rather than direct translations or simple repackaging, allowing the benchmark to reflect phenomena and usage patterns specific to the language." (p.5)
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)

Strengths

- All four deployment-required tasks (HA, LA, TC, NER) are present in the benchmark [Q15, Q29, Q46, Q34]
- Headline acceptability is purpose-built for Luxembourgish rather than ported [Q69], directly serving the deployment's headline-consistency requirement
- Topic taxonomy maps to standard international news categories, matching the deployment's stated taxonomy needs [Q48, A1]

ID	Evidence
Q15	"We formulate headline acceptability (HA) as a binary classification task where the model must decide whether a given headline matches the accompanying article body." (p.3)
Q29	"We introduce a linguistic acceptability dataset consisting of four distinct linguistic subtypes, which can either be used as a binary (LA (BINARY)) or multiclass (LA (MULTI)) classification dataset." (p.3)
Q34	"The JUDGEWEL dataset (Plum et al., 2026) introduces an automatically constructed corpus for named entity recognition (NER) in LTZ, derived from Wikipedia and Wikidata." (p.4)

Q46	"To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories." (p.4)
Q48	"From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS." (p.4)
Q69	"In addition, a substantial proportion of the LTZGLUE tasks are newly created for LTZ rather than direct translations or simple repackaging, allowing the benchmark to reflect phenomena and usage patterns specific to the language." (p.5)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Deployment requires HA, LA, TC, NER tasks under standard international news categories with code-switching robustness [A1, A2]; all four task types are represented in LTZGLUE [Q15, Q29, Q46, Q34].

IO-2: The TC label set (SPORTS, CULTURE, TECHNOLOGY, BUSINESS, ANIMALS) [Q48] omits standard newsroom beats such as POLITICS, HEALTH, JUSTICE, and LOCAL/MUNICIPAL, which are core to Luxembourgish journalism including EU institutional coverage [WEB-1]. No headline-acceptability category gap; HA is present [Q15].

IO-3: Three benchmark tasks (Sentiment Analysis [Q23], Intent Detection [Q51, Q154], Recognising Textual Entailment [Q59]) are not part of the deployment use case and would consume evaluation effort without contributing to deployment validity. The ANIMALS topic label [Q48] is an unusual inclusion whose relevance to a professional newsroom taxonomy is not justified in the documentation.

IO-4: Category gaps within topic classification (missing POLITICS/HEALTH/JUSTICE) and inclusion of ANIMALS create content-validity concerns specifically for the TC task; the HA, LA, and NER task categories themselves are appropriate for the deployment.

ID	Evidence
Q15	"We formulate headline acceptability (HA) as a binary classification task where the model must decide whether a given headline matches the accompanying article body." (p.3)
Q23	"We formulate the sentiment analysis (SA) task as a classification task where the model has to predict positive, negative, and neutral sentiment." (p.3)
Q29	"We introduce a linguistic acceptability dataset consisting of four distinct linguistic subtypes, which can either be used as a binary (LA (BINARY)) or multiclass (LA (MULTI)) classification dataset." (p.3)
Q34	"The JUDGEWEL dataset (Plum et al., 2026) introduces an automatically constructed corpus for named entity recognition (NER) in LTZ, derived from Wikipedia and Wikidata." (p.4)
Q46	"To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories." (p.4)
Q48	"From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS." (p.4)
Q51	"We constructed a new LTZ dataset for intent detection (ID) by translating the English xSID test and validation datasets (van der Goot et al., 2021)." (p.4)
Q59	"Recognizing Textual Entailment (RTE) (Haim et al., 2006) is a classic NLU task featured in the original GLUE benchmark. Given a pair of texts A and B, the task consists of determining whether A is a logical premise of B." (p.5)
Q154	"slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm." (p.15)

WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)
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Information Gaps

- No documentation of why ANIMALS was selected as a top-five news category; whether RTL's editorial taxonomy actually contains a comparable label is not addressed
- No documentation of category coverage for POLITICS / EU affairs in the TC training set despite their centrality to Luxembourgish journalism

Requires Expert Verification

- Whether the deploying outlet's editorial taxonomy actually requires only the five benchmark TC labels or whether additional Luxembourg-specific beats are operationally needed

Remediation

Gap 1: TC label set (SPORTS, CULTURE, TECHNOLOGY, BUSINESS, ANIMALS) [Q48] omits standard newsroom beats (POLITICS, HEALTH, JUSTICE, LOCAL/MUNICIPAL) and includes ANIMALS without operational justification

Recommendation: Audit the deploying outlet's actual editorial taxonomy and either retrain TC against that taxonomy or document mappings/exclusions; verify whether ANIMALS aligns with any operational category

ID	Evidence
Q48	"From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS." (p.4)

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Code-switching is identified as a HIGH-priority deployment requirement [A2] and is a structural feature of Luxembourgish journalism [WEB-6, WEB-10]. The benchmark explicitly acknowledges that 'most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts' [Q112], and that data are concentrated in a limited set of public domains [Q119]. Only the human-annotated RTL news comment NER corpus is documented as covering 'informal and code-mixed writing' [Q41]; systematic quantification of code-switched content across the other tasks (HA, LA, TC) is absent. RTL is the principal source for HA, SA, TC, and NER comments [Q16, Q24, Q40, Q46], which aligns with the deployment's primary outlet [WEB-1], a meaningful strength. However, LA sentences are sourced from the LOD formal dictionary [Q30], RTE is machine-translated then LLM-improved [Q60, Q61], and intent detection uses German MT [Q54] — meaning much of the benchmark content is not authentic Luxembourgish journalistic text. Given code-switching is a HIGH-priority dimension for this deployment and the benchmark itself flags this as a limitation, score is 2.

ID	Evidence
Q16	"To construct this dataset, we use RTL news articles. We keep only documents from the twenty most frequent categories. We then filter articles by body length and title length, remove exact duplicate titles, randomly shuffle the remaining instances, and retain a fixed subset of 30k examples." (p.3)
Q24	"We use articles from RTL, randomly selected from the commentary and letter to the editor sections." (p.3)
Q30	"The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset." (p.3)
Q40	"The NER dataset introduced by Lothritz et al. (2022), by contrast, is a fully human-annotated corpus derived from RTL online news comments." (p.4)
Q41	"It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE)." (p.4)
Q46	"To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories." (p.4)
Q54	"Since this task is originally intended to be crosslingual, we use the machine translated German training set (van der Goot et al., 2021)." (p.5)
Q60	"Lothritz et al. (2023) released a machine-translated Luxembourgish version of the dataset using Google Translate. However, due to numerous grammar and vocabulary related mistakes introduced in this process, we set out to improve the quality of the dataset." (p.5)
Q61	"Specifically, we first prompted CHATGPT-5.1 to assess and improve the translated sentence pairs unless they were already of very high quality, while explicitly keeping the original meaning to avoid label conflicts (see Appendix 7.4)." (p.5)
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)
Q119	"LTZ is a small language community, and linguistic data often originate from a limited set of public domains." (p.10)
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)
WEB-6	Luxembourg's three-language administrative regime (Law of 24 February 1984) institutionalises French/German/Luxembourgish co-occurrence in journalistic text (https://en.wikipedia.org/wiki/Languages_of_Luxembourg)

WEB-10	Luxembourgish nationals speak on average 4.3 languages and 9 of 10 residents speak French, confirming code-switching as a structural feature (https://en.paperjam.lu/article/delano_23-lux-residents-speak-4-or-more-languages)
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Strengths

- RTL — the principal Luxembourgish broadcast outlet [WEB-1] — is the dominant source for HA, SA, TC, and NER, providing strong domain alignment for the deployment's primary register [Q16, Q24, Q40, Q46]
- The RTL news comment NER subcorpus explicitly covers informal and code-mixed writing [Q41], partially addressing the deployment's code-switching need for NER
- The benchmark explicitly acknowledges its register and code-switching limitations [Q112, Q119], enabling informed deployment risk management

ID	Evidence
Q16	"To construct this dataset, we use RTL news articles. We keep only documents from the twenty most frequent categories. We then filter articles by body length and title length, remove exact duplicate titles, randomly shuffle the remaining instances, and retain a fixed subset of 30k examples." (p.3)
Q24	"We use articles from RTL, randomly selected from the commentary and letter to the editor sections." (p.3)
Q40	"The NER dataset introduced by Lothritz et al. (2022), by contrast, is a fully human-annotated corpus derived from RTL online news comments." (p.4)
Q41	"It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE)." (p.4)
Q46	"To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories." (p.4)
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)
Q119	"LTZ is a small language community, and linguistic data often originate from a limited set of public domains." (p.10)
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — Luxembourgish journalism requires handling French, German, and English code-switching, EU institutional terminology, and Grande Région cross-border entities [A2, WEB-1, WEB-10]. The benchmark partially addresses this through the RTL comment NER subcorpus [Q41] but does not systematically cover code-switching across HA, LA, or TC.

IC-2: RTL-derived content (HA, SA, TC, NER comments) [Q16, Q24, Q40, Q46] aligns with mainstream Luxembourgish journalistic culture; LA sentences from LOD [Q30] reflect formal/normative register diverging from the deployment's flexible journalistic standard [A3].

IC-3: Intent detection uses xSID voice-assistant commands [Q51, Q55] — a register entirely irrelevant to newsroom deployment. RTE is sourced from English-origin pairs translated and LLM-improved [Q60, Q61], introducing

non-Luxembourgish provenance.

IC-4: INSUFFICIENT DOCUMENTATION — no regional annotator audit of code-switched coverage across the benchmark has been performed; gap_id 1 in the regional YAML remains unresolved [NEEDS VERIFICATION]. Direct dataset inspection by Luxembourgish journalism experts would be needed.

IC-5: Documented content issues: (a) systematic under-representation of code-switching outside the NER comment subcorpus [Q112]; (b) LA grounded in formal LOD register [Q30] rather than journalistic register; (c) 22–28% of RTE filtered post-LLM improvement [Q66] indicating quality variance; (d) JUDGEWEL automatically constructed with minimal human verification [Q37].

ID	Evidence
Q16	“To construct this dataset, we use RTL news articles. We keep only documents from the twenty most frequent categories. We then filter articles by body length and title length, remove exact duplicate titles, randomly shuffle the remaining instances, and retain a fixed subset of 30k examples.” (p.3)
Q24	“We use articles from RTL, randomly selected from the commentary and letter to the editor sections.” (p.3)
Q30	“The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset.” (p.3)
Q37	“The resulting sentences are then evaluated using LLMs acting as judges, with minimal human verification to calibrate quality thresholds.” (p.4)
Q40	“The NER dataset introduced by Lothritz et al. (2022), by contrast, is a fully human-annotated corpus derived from RTL online news comments.” (p.4)
Q41	“It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE).” (p.4)
Q46	“To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories.” (p.4)
Q51	“We constructed a new LTZ dataset for intent detection (ID) by translating the English xSID test and validation datasets (van der Goot et al., 2021).” (p.4)
Q55	“The main challenge in translating the English dataset stems from its register. The source segments consist of user commands for a voice-controlled AI assistant, representing a specialised spoken register for which there is no equivalent reference corpus in LTZ. This register is marked by domain-specific terminology and collocations (e.g., set an alarm, set a reminder, add to playlist), as well as non-standard spelling (e.g., all lowercase, missing punctuation).” (p.5)
Q60	“Lothritz et al. (2023) released a machine-translated Luxembourgish version of the dataset using Google Translate. However, due to numerous grammar and vocabulary related mistakes introduced in this process, we set out to improve the quality of the dataset.” (p.5)
Q61	“Specifically, we first prompted CHATGPT-5.1 to assess and improve the translated sentence pairs unless they were already of very high quality, while explicitly keeping the original meaning to avoid label conflicts (see Appendix 7.4).” (p.5)
Q66	“The filtering reduced between 22 and 28% of instances in the data, resulting in a final dataset of 1,876, 197, and 626 sentence pairs for the training, development, and test set, respectively.” (p.5)
Q112	“Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts.” (p.9)
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)
WEB-6	Luxembourg's three-language administrative regime (Law of 24 February 1984) institutionalises French/German/Luxembourgish co-occurrence in journalistic text (https://en.wikipedia.org/wiki/Languages_of_Luxembourg)
WEB-10	Luxembourgish nationals speak on average 4.3 languages and 9 of 10 residents speak French, confirming code-switching as a structural feature (https://en.paperjam.lu/article/delano_23-lux-residents-speak-4-or-more-languages)

Information Gaps

- Quantified rate of French/German/English code-switching in HA, LA, and TC instances (gap_id 1, deferred)
- Whether RTL article corpus reflects only published staff articles or includes user-comment register
- Coverage of EU institutional naming conventions and Grande Région entity mentions in the NER training set

Requires Expert Verification

- Whether code-switching density in benchmark instances matches operational density in target newsroom copy
 - Whether the RTL-sourced subset reflects the same editorial register as the deploying outlet's output
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Remediation

Gap 1: Code-switching density across HA, LA, and TC tasks is undocumented; only the NER comment subcorpus explicitly covers code-mixed text [Q41, Q112]

Recommendation: Sample 200–500 instances per task and label code-switched tokens; commission a code-switched evaluation supplement drawn from RTL articles to quantify performance under realistic French/German/English mixing

ID	Evidence
Q41	"It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE)." (p.4)
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Modality alignment is good: both benchmark and deployment are text-only [Q17], RTL articles, comments, parliamentary transcripts, and chat rooms provide register diversity in pre-training [Q74], and OpenLID filtering removes non-Luxembourgish content [Q47]. The 40–400 word filter [Q47] is plausibly compatible with newsroom article lengths. However, Luxembourgish has 'ongoing standardisation' and 'vast amounts of variation' [Q7], and the documentation does not describe explicit orthographic normalisation across datasets. The BPE tokenizer (50,368 vocab on 233M tokens) [Q128, Q75] may not adequately represent journalistic orthographic variants or code-switched French/German tokens, particularly given the formal-register skew of the data [Q112]. The deployment explicitly applies the flexible journalistic register [A3], but LA Ortho ground truth is built from Spellchecker.lu data [Q31] encoding the formal ZLS standard [WEB-11, WEB-12], creating a known register mismatch.

ID	Evidence
Q7	“Additional factors, such as the ongoing standardisation of the language (Gilles, 2019), vast amounts of variation (Lutgen et al., 2025), and decentralised resources, make it extremely challenging to evaluate LTZ language understanding in language models.” (p.1)
Q17	“This subset is split equally, with one half serving as the positive class with original headlines, and the other half providing the article bodies for which we assign swapped headlines.” (p.3)
Q31	“The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data.” (p.3)
Q47	“We applied a series of preprocessing steps to ensure data quality. Specifically, we removed articles identified as non-Luxembourgish by OpenLID (Burchell et al., 2023), as well as those containing fewer than 40 words or more than 400 words.” (p.4)
Q74	“The pre-training set is compiled from a variety of sources of LTZ. A large portion of the data stems from RTL (see Section 3), including news articles (News), transcribed radio interviews (Radio), and user comments (Comments). We also include transcribed podcasts (Podcasts) and transcribed political speeches and debates from the Chambre des Députés (Chamber). In addition, we use 1M sentences from the web crawl of the Leipzig Collection (Web, this excludes RTL), text crawled from LTZ chat rooms (Webchat), a Wikipedia crawl from October 2023 (Wikipedia), and finally, example sentences from the LOD retrieved in March 2024.” (p.6)
Q75	“We filter out sentences containing fewer than three words (as tokenized by whitespace), totalling 11.7M sentences, which corresponds to roughly 233M tokens using our tokenizer.” (p.6)
Q112	“Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts.” (p.9)
Q128	“Both models share a GPTNeoXTokenizerFast tokenizer (Black et al., 2022), a BPE-based tokenizer, which we train on the entire pre-training set, using a minimum frequency of two and a vocabulary size of 50,368.” (p.12)
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies (https://data.public.lu/en/organizations/zenter-fir-dletzebuerger-sprooch/)
WEB-12	https://men.public.lu/en/grands-dossiers/systeme-educatif/promotion-langue-luxembourgeoise.html

Strengths

- Text-only modality matches the deployment's text-only pipeline [Q17]
- Pre-training corpus draws from a broad set of Luxembourgish written sources including RTL news, comments, parliamentary speech, podcasts, chat rooms, and Wikipedia [Q74], providing some register breadth
- Article-length filtering (40–400 words) [Q47] is operationally compatible with newsroom article lengths

ID	Evidence
Q17	"This subset is split equally, with one half serving as the positive class with original headlines, and the other half providing the article bodies for which we assign swapped headlines." (p.3)
Q47	"We applied a series of preprocessing steps to ensure data quality. Specifically, we removed articles identified as non-Luxembourgish by OpenLID (Burchell et al., 2023), as well as those containing fewer than 40 words or more than 400 words." (p.4)
Q74	"The pre-training set is compiled from a variety of sources of LTZ. A large portion of the data stems from RTL (see Section 3), including news articles (News), transcribed radio interviews (Radio), and user comments (Comments). We also include transcribed podcasts (Podcasts) and transcribed political speeches and debates from the Chambre des Députés (Chamber). In addition, we use 1M sentences from the web crawl of the Leipzig Collection (Web, this excludes RTL), text crawled from LTZ chat rooms (Webchat), a Wikipedia crawl from October 2023 (Wikipedia), and finally, example sentences from the LOD retrieved in March 2024." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both benchmark and deployment use plain text [Q17]; signal-distribution mismatch is low at the modality level. Tokenisation (BPE, 50,368 vocab on 233M tokens) [Q128, Q75] may under-represent journalistic orthographic variants and code-switched tokens.

IF-2: Luxembourg has near-universal digital infrastructure (98.8% internet penetration [WEB-17]), so capture infrastructure is not a constraint.

IF-3: The benchmark filters articles to 40–400 words [Q47] and removes non-Luxembourgish content via OpenLID [Q47] — both operationally plausible for newsroom text. However, no explicit orthographic normalisation across datasets is documented despite acknowledged 'vast amounts of variation' [Q7], and the LA Ortho category encodes the ZLS formal standard via Spellchecker.lu [Q31, WEB-11] which diverges from the deployment's flexible journalistic register [A3].

IF-4: Form-level mismatches that may harm external validity: (a) tokenisation potentially mis-segmenting code-switched and orthographically variant tokens; (b) implicit assumption of formal orthography in LA Ortho ground truth diverging from journalistic practice [A3, Q31].

ID	Evidence
Q7	"Additional factors, such as the ongoing standardisation of the language (Gilles, 2019), vast amounts of variation (Lutgen et al., 2025), and decentralised resources, make it extremely challenging to evaluate LTZ language understanding in language models." (p.1)
Q17	"This subset is split equally, with one half serving as the positive class with original headlines, and the other half providing the article bodies for which we assign swapped headlines." (p.3)
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)

Q47	"We applied a series of preprocessing steps to ensure data quality. Specifically, we removed articles identified as non-Luxembourgish by OpenLID (Burchell et al., 2023), as well as those containing fewer than 40 words or more than 400 words." (p.4)
Q75	"We filter out sentences containing fewer than three words (as tokenized by whitespace), totalling 11.7M sentences, which corresponds to roughly 233M tokens using our tokenizer." (p.6)
Q128	"Both models share a GPTNeoXTokenizerFast tokenizer (Black et al., 2022), a BPE-based tokenizer, which we train on the entire pre-training set, using a minimum frequency of two and a vocabulary size of 50,368." (p.12)
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies (https://data.public.lu/en/organizations/zenter-fir-dletzebuerger-sprooch/)
WEB-17	98.8% internet penetration in Luxembourg confirms enterprise-grade digital infrastructure for newsroom deployment (https://www.theglobaleconomy.com/Luxembourg/Internet_users/)

Information Gaps

- Whether explicit orthographic normalisation was applied across datasets (gap_id 3, partially resolved — formal-source inputs confirmed but normalisation across datasets undocumented)
- Tokenizer behaviour on code-switched and orthographically variant tokens

Requires Expert Verification

- Whether the BPE tokenizer adequately segments code-switched newsroom copy

Remediation

Gap 1: No explicit orthographic normalisation across datasets despite acknowledged 'vast amounts of variation' [Q7]; tokenizer behaviour on code-switched and orthographically variant tokens is undocumented [Q128]

Recommendation: Profile the BPE tokenizer's handling of code-switched French/German tokens and journalistic orthographic variants on a sample of newsroom copy; consider tokenizer extension or fine-tuning if fragmentation rates are high

ID	Evidence
Q7	"Additional factors, such as the ongoing standardisation of the language (Gilles, 2019), vast amounts of variation (Lutgen et al., 2025), and decentralised resources, make it extremely challenging to evaluate LTZ language understanding in language models." (p.1)
Q128	"Both models share a GPTNeoXTokenizerFast tokenizer (Black et al., 2022), a BPE-based tokenizer, which we train on the entire pre-training set, using a minimum frequency of two and a vocabulary size of 50,368." (p.12)

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output label structure is appropriate for the deployment's per-task classification paradigm [Q84, A4], and headline acceptability binary (True/False) [Q148] and TC's category labels [Q153] map cleanly to editorial use. However, several output-ontology features misalign with deployment needs (HIGH priority): (a) LA Multi exposes formally-derived linguistic error categories (Verb, Adj, Syntax, Ortho) [Q31, Q151] grounded in the LOD/Spellchecker formal standard, while the deployment explicitly applies a flexible journalistic register [A3] — the Ortho category in particular may penalise legitimate journalistic variants; (b) the GPE→LOC merge [Q44] reduces granularity for cross-border journalism that needs to distinguish municipalities, regions, and countries [WEB-1, regional YAML]; (c) the ANIMALS topic label [Q48] encodes a non-newsroom assumption; (d) intent detection's xSID schema [Q154] is operationally irrelevant. The benchmark explicitly cautions against using performance as evidence of cultural/demographic coverage [Q121].

ID	Evidence
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)
Q44	"To ensure compatibility, the tag set is harmonised by merging the GPE and LOC categories into a single location label, while retaining PER, ORG, and MISC unchanged." (p.4)
Q48	"From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS." (p.4)
Q84	"We did not use a Multiple Choice Question Answering (MCQA)-setup, but provided the labels that should be used as output." (p.6)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)
Q148	"headline_classification: Decide if the given title/headline fits the text. Output True or False." (p.15)
Q151	"linguistic_acceptability_multilabel: Detect if the sentence is correct or if some element is wrong. If the sentence is correct, Output: correct. If it is not, Output the label referencing the wrong element: syntax, verb, ortho or adj." (p.15)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)
Q154	"slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm." (p.15)
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)

Strengths

- Label-output (not MCQA) paradigm [Q84] mirrors the deployment's per-task classification with valid label sets, matching the deployment's per-task scoring pipeline [A4]
- HA binary True/False [Q148] directly maps to the deployment's headline-flagging logic

- TC label set covers standard international news categories aligned with the deployment's editorial taxonomy [Q153, A1]

ID	Evidence
Q84	"We did not use a Multiple Choice Question Answering (MCQA)-setup, but provided the labels that should be used as output." (p.6)
Q148	"headline_classification: Decide if the given title/headline fits the text. Output True or False." (p.15)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: HA [Q148] and TC [Q153] labels align with deployment needs. NER tags (LOC/PER/ORG/MISC/DATE) [Q152] are appropriate categories but the GPE→LOC merge [Q44] is regionally suboptimal. LA Multi categories [Q151] encode formal linguistic error types divergent from the deployment's journalistic register [A3].

OO-2: GPE has been merged into LOC [Q44], removing a category Luxembourgish cross-border journalism plausibly needs to distinguish (municipalities vs. countries vs. geographic regions), per regional YAML and [WEB-1]. No journalist-register acceptability category exists.

OO-3: The LA Ortho category [Q31, Q151] encodes Spellchecker.lu's formal ZLS standard [WEB-11], which represents non-deployment values (formal orthographic correctness) rather than the editorial register the deployment applies [A3]. The ANIMALS TC label [Q48] reflects an editorial assumption inherited from RTL's category structure rather than universally newsroom-relevant.

OO-4: Stakeholder-driven taxonomy redesign is warranted for LA Multi (to recognise journalistically-acceptable variants) and for TC (to evaluate whether ANIMALS is needed and whether POLITICS / EU coverage labels should be added) given HIGH OO priority weight.

OO-5: Documented taxonomy issues: (a) LA Ortho derived from formal-standard Spellchecker.lu [Q31] vs. flexible journalistic register [A3]; (b) GPE→LOC merge [Q44] vs. cross-border granularity needs; (c) ANIMALS label [Q48] vs. standard newsroom beats; (d) intent detection schema [Q154] irrelevant to deployment.

ID	Evidence
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)
Q44	"To ensure compatibility, the tag set is harmonised by merging the GPE and LOC categories into a single location label, while retaining PER, ORG, and MISC unchanged." (p.4)
Q48	"From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS." (p.4)
Q148	"headline_classification: Decide if the given title/headline fits the text. Output True or False." (p.15)
Q151	"linguistic_acceptability_multilabel: Detect if the sentence is correct or if some element is wrong. If the sentence is correct, Output: correct. If it is not, Output the label referencing the wrong element: syntax, verb, ortho or adj." (p.15)

Q152	"ner: Perform Named Entity Recognition on the given sequence of sentence tokens. Output tags as lists of ner_tags. Allowed Tags: O, B-LOC, I-LOC, B-PER, I-PER, B-DATE, I-DATE, B-ORG, I-ORG, B-MISC, I-MISC." (p.15)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)
Q154	"slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm." (p.15)
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies (https://data.public.lu/en/organizations/zenter-fir-dletzebuerger-sprooch/)

Information Gaps

- Whether LA Ortho ground-truth labels would survive review by professional Luxembourgish editors applying journalistic register
- Whether the deploying outlet's NER schema requires the GPE/LOC distinction

Requires Expert Verification

- Editorial review of a sample of LA Ortho instances against journalistic-register standards
- Editorial confirmation of the operational TC taxonomy

Remediation

Gap 1: LA Ortho ground truth derived from Spellchecker.lu / formal ZLS standard [Q31, WEB-11] diverges from the deployment's flexible journalistic register [A3]

Recommendation: Either drop the Ortho subclass for production scoring or remap it against an editorial style guide; involve the deploying outlet's chief sub-editor in defining what counts as an unacceptable orthographic variant

Gap 2: GPE has been merged into LOC [Q44], reducing granularity for Luxembourgish cross-border journalism that needs to distinguish municipalities, regions, countries, and EU institutions [WEB-1]

Recommendation: Restore the GPE/LOC distinction in a deployment-specific NER schema and supplement training data with annotated EU institutional and Grande Région entity examples

ID	Evidence
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)
Q44	"To ensure compatibility, the tag set is harmonised by merging the GPE and LOC categories into a single location label, while retaining PER, ORG, and MISC unchanged." (p.4)
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies (https://data.public.lu/en/organizations/zenter-fir-dletzebuerger-sprooch/)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotator-population mismatch is the highest-confidence concern in this analysis. Sentiment analysis annotation achieved Cohen's Kappa of only 0.45 [\[Q27\]](#) — below the 0.60–0.80 range typical for robust NLU annotation — with disagreements resolved by annotators agreeing among themselves [\[Q28\]](#). Annotators beyond SA are described as 'student assistants' [\[Q114\]](#); no evidence of professional journalist or editor involvement, despite the deployment explicitly requiring journalistic-register ground truth [\[A3\]](#). RTE labels are AI-assisted (ChatGPT-5.1 improvement [\[Q61\]](#), ChatGPT-5-MINI quality and label verification [\[Q63, Q64\]](#)), with manual correction only where the LLM altered semantics [\[Q65\]](#). JUDGEWEL NER uses LLMs as judges with minimal human verification [\[Q37\]](#). The benchmark itself warns 'models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices' [\[Q120\]](#). Given OC is a HIGH-priority dimension and the deployment requires journalist-register ground truth, score is 2.

ID	Evidence
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q28	"For the final set, the annotators agree on a label in cases of label disagreement." (p.3)
Q37	"The resulting sentences are then evaluated using LLMs acting as judges, with minimal human verification to calibrate quality thresholds." (p.4)
Q61	"Specifically, we first prompted CHATGPT-5.1 to assess and improve the translated sentence pairs unless they were already of very high quality, while explicitly keeping the original meaning to avoid label conflicts (see Appendix 7.4)." (p.5)
Q63	"To achieve (a), we prompted CHATGPT-5-MINI to judge the texts in the improved data and label their quality as either low, medium, or high, keeping only data rated at least medium, removing nearly 25% of the entire dataset (see Appendix 7.5)." (p.5)
Q64	"For (b), we prompted CHATGPT-5-MINI to verify whether the dataset labels remained correct after the first translation and improvement, outputting true or false for each sentence pair (see Appendix 7.6). Nearly 10% of the labels were false." (p.5)
Q65	"We found that the quality improvement step often corrected intentional logical contradictions or factual inaccuracies rather than keeping the original semantics. We therefore adjusted the sentences manually such that they corresponded to the ground truth again, while keeping false positives intact." (p.5)
Q114	"We would like to thank the student assistants for their annotation work." (p.9)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)

Strengths

- The Lothritz et al. (2022) RTL NER subcorpus is fully human-annotated and high-precision [\[Q40, Q42\]](#)
- RTE labels underwent two-stage AI verification with manual correction where semantics drifted [\[Q62, Q65\]](#), offering some quality control
- The benchmark explicitly cautions deployers against treating performance as cultural/demographic coverage [\[Q121\]](#), enabling informed risk assessment

ID	Evidence
Q40	"The NER dataset introduced by Lothritz et al. (2022), by contrast, is a fully human-annotated corpus derived from RTL online news comments." (p.4)
Q42	"Annotation was conducted manually, yielding a smaller but high-precision dataset." (p.4)

Q62	"In addition, we perform two verification steps to make sure that (a) the quality of the improved texts is high enough and (b) that the labels are correct." (p.5)
Q65	"We found that the quality improvement step often corrected intentional logical contradictions or factual inaccuracies rather than keeping the original semantics. We therefore adjusted the sentences manually such that they corresponded to the ground truth again, while keeping false positives intact." (p.5)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels do not reflect professional Luxembourgish journalist/editor perspectives. The deployment requires the flexible journalistic register [A3], but annotators are described as student assistants [Q114] with no documented journalism background; LA labels derive from formal LOD/Spellchecker sources [Q30, Q31] and RTE labels are AI-assisted [Q63, Q64].

OC-2: Likely substantial disagreement between benchmark annotators and the target population (working journalists/editors), particularly on borderline LA Ortho cases [A3] where the deployment explicitly anticipates formal-vs-journalistic disagreement.

OC-3: Annotator demographics: SA was annotated by two native Luxembourgish speakers [Q25] with Cohen's Kappa 0.45 [Q27]; intent translation by one native speaker with consultation [Q52]; other tasks by 'student assistants' [Q114]. No professional-journalist annotators documented; no Datasheet-level demographic breakdown provided.

OC-4: Re-annotation of LA, HA, and a sample of NER by professional Luxembourgish editors applying the deployment's flexible register would be necessary to establish convergent validity for the deployment.

OC-5: SA disagreements resolved by annotators agreeing among themselves [Q28] — a consensus aggregation that erases minority annotator perspectives without documented arbitration. Cohen's Kappa 0.45 [Q27] indicates substantial baseline disagreement subsequently flattened.

OC-6: Documented label issues likely to harm convergent and external validity: (a) student-annotator pool [Q114] vs. professional-journalist target users [A3]; (b) AI-assisted RTE labels [Q63, Q64]; (c) LLM-as-judge JUDGEWEL [Q37]; (d) low SA IAA [Q27]; (e) explicit benchmark caution against demographic-coverage claims [Q120, Q121].

ID	Evidence
Q25	"In total, we extract 4,583 sentences, which are then annotated by two native speakers of LTZ." (p.3)
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q28	"For the final set, the annotators agree on a label in cases of label disagreement." (p.3)
Q30	"The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset." (p.3)
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)

Q37	"The resulting sentences are then evaluated using LLMs acting as judges, with minimal human verification to calibrate quality thresholds." (p.4)
Q52	"The translations were performed by an LTZ native speaker. In cases of uncertainty, additional native LTZ speakers were consulted." (p.4)
Q63	"To achieve (a), we prompted CHATGPT-5-MINI to judge the texts in the improved data and label their quality as either low, medium, or high, keeping only data rated at least medium, removing nearly 25% of the entire dataset (see Appendix 7.5)." (p.5)
Q64	"For (b), we prompted CHATGPT-5-MINI to verify whether the dataset labels remained correct after the first translation and improvement, outputting true or false for each sentence pair (see Appendix 7.6). Nearly 10% of the labels were false." (p.5)
Q114	"We would like to thank the student assistants for their annotation work." (p.9)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies (https://data.public.lu/en/organizations/zenter-fir-dletzebuerger-sprooch/)

Information Gaps

- Number, demographics, and journalism background of student annotators
- Whether any IAA was computed for tasks other than SA
- Adjudication procedure when SA annotators initially disagreed

Requires Expert Verification

- Disagreement rate between professional Luxembourgish editors and benchmark labels on a held-out sample, especially for LA Ortho and HA

Remediation

Gap 1: Annotators are student assistants [Q114] with no professional journalism background; SA Cohen's Kappa 0.45 [Q27] is below typical thresholds; RTE and JUDGEWEL labels are AI-assisted [Q37, Q63, Q64]

Recommendation: Re-annotate held-out HA and LA test sets with at least three professional Luxembourgish editors applying explicit journalistic-register guidelines; report IAA and adjudication procedure

ID	Evidence
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q37	"The resulting sentences are then evaluated using LLMs acting as judges, with minimal human verification to calibrate quality thresholds." (p.4)
Q63	"To achieve (a), we prompted CHATGPT-5-MINI to judge the texts in the improved data and label their quality as either low, medium, or high, keeping only data rated at least medium, removing nearly 25% of the entire dataset (see Appendix 7.5)." (p.5)
Q64	"For (b), we prompted CHATGPT-5-MINI to verify whether the dataset labels remained correct after the first translation and improvement, outputting true or false for each sentence pair (see Appendix 7.6). Nearly 10% of the labels were false." (p.5)
Q114	"We would like to thank the student assistants for their annotation work." (p.9)

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form aligns well with deployment needs (LOWER priority dimension). All tasks produce discrete label outputs evaluated by macro-F1 [\[Q93, Q96\]](#), directly compatible with the deployment's per-task significance-score-and-threshold pipeline [\[A4\]](#). Encoder results are averaged over three runs with reported standard deviations [\[Q89, Q96\]](#), enabling stability assessment. The label-output (not MCQA) paradigm [\[Q84\]](#) mirrors the deployment's classification framing. Class-imbalanced tasks use class-balanced loss [\[Q92\]](#). Caveats: LLM outputs failing label format are discarded [\[Q90\]](#), inflating reported scores; LLMs are evaluated only once with macro-F1 only [\[Q96\]](#) — meaning LLM performance is less stable, and the benchmark itself flags this as 'indicative rather than directly comparable' [\[Q86\]](#). The deployment's threshold-flagging logic [\[A4\]](#) requires confidence scores per task, which encoder fine-tuning supports natively but prompted LLM evaluation may not.

ID	Evidence
Q84	"We did not use a Multiple Choice Question Answering (MCQA)-setup, but provided the labels that should be used as output." (p.6)
Q86	"However, prompt-based evaluation introduces additional sources of variability, including prompt sensitivity and differences in instruction-following behaviour across models. As a result, performance should be interpreted as indicative rather than directly comparable to supervised results." (p.6)
Q89	"For encoder-based models, results are reported as averages over multiple runs (see Appendix 7.2 for more details)." (p.7)
Q90	"Prompted LLMs do not always produce well-formed outputs and may return an incorrect number of predictions for a given task; such outputs are discarded prior to evaluation." (p.7)
Q92	"For the supervised models, since the linguistic acceptability and sentiment analysis datasets are highly imbalanced, when fine-tuning on these tasks we use class-balanced loss based on effective size (Cui et al., 2019) with a beta of 0.99." (p.7)
Q93	"Table 6 shows F1 scores for all models across all tasks (see Appendix 7.9 for full results)." (p.7)
Q96	"Table 6: Test F1 scores across all ItzGLUE tasks. Encoder results are averaged over three runs with standard deviations as subscripts. Prompted LLMs were evaluated once; we report macro-F1 only." (p.8)

Strengths

- Discrete label outputs evaluated by macro-F1 [\[Q93, Q96\]](#) directly support the deployment's per-task threshold flagging [\[A4\]](#)
- Encoder results averaged over three runs with standard deviations [\[Q89, Q96\]](#) permit stability assessment for production deployment
- Class-balanced loss applied to imbalanced tasks (LA, SA) [\[Q92\]](#) addresses real distributional skew
- Label-output paradigm rather than MCQA [\[Q84\]](#) matches the deployment's classification framing

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Q84	"We did not use a Multiple Choice Question Answering (MCQA)-setup, but provided the labels that should be used as output." (p.6)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Discrete label outputs [Q93, Q148, Q149, Q150, Q151, Q152, Q153] match the deployment's per-task scoring needs [A4]; the deployment's threshold flagging is a downstream wrapper compatible with macro-F1 evaluation.

OF-2: Not applicable — deployment is text-only [deployment_modality]; speech output is not required.

OF-3: Target users are professional journalists/editors with high literacy and trilingual proficiency [WEB-14, WEB-15]; no accessibility constraints introduced by the output form.

OF-4: Minor mismatches: (a) prompted LLM scores are single-run macro-F1 [Q96] less directly comparable to encoder runs [Q86]; (b) discarded malformed LLM outputs [Q90] inflate apparent performance — relevant if the deployment uses prompted LLMs rather than fine-tuned encoders.

ID	Evidence
Q86	"However, prompt-based evaluation introduces additional sources of variability, including prompt sensitivity and differences in instruction-following behaviour across models. As a result, performance should be interpreted as indicative rather than directly comparable to supervised results." (p.6)
Q90	"Prompted LLMs do not always produce well-formed outputs and may return an incorrect number of predictions for a given task; such outputs are discarded prior to evaluation." (p.7)
Q93	"Table 6 shows F1 scores for all models across all tasks (see Appendix 7.9 for full results)." (p.7)
Q96	"Table 6: Test F1 scores across all ItzGLUE tasks. Encoder results are averaged over three runs with standard deviations as subscripts. Prompted LLMs were evaluated once; we report macro-F1 only." (p.8)
Q148	"headline_classification: Decide if the given title/headline fits the text. Output True or False." (p.15)
Q149	"sentiment_analysis: Classify sentiment of the text. Allowed labels: positive, neutral, negative." (p.15)
Q150	"linguistic_acceptability_binary: Decide whether the sentence is linguistically acceptable in Luxembourgish. Output: 0 or 1." (p.15)
Q151	"linguistic_acceptability_multilabel: Detect if the sentence is correct or if some element is wrong. If the sentence is correct, Output: correct. If it is not, Output the label referencing the wrong element: syntax, verb, ortho or adj." (p.15)
Q152	"ner: Perform Named Entity Recognition on the given sequence of sentence tokens. Output tags as lists of ner_tags. Allowed Tags: O, B-LOC, I-LOC, B-PER, I-PER, B-DATE, I-DATE, B-ORG, I-ORG, B-MISC, I-MISC." (p.15)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)
WEB-14	Luxembourg trilingual education system produces newsroom professionals with high literacy in Luxembourgish/French/German plus English (https://men.public.lu/en/systeme-educatif/langues-ecole-luxembourgaise.html)
WEB-15	EU Education and Training Monitor confirms strong language education context for target users (https://op.europa.eu/webpub/eac/education-and-training-monitor/en/country-reports/luxembourg.html)

Information Gaps

- Whether confidence scores from fine-tuned encoders are well-calibrated for the deployment's threshold-based flagging logic

Requires Expert Verification

- Calibration of model confidence outputs against editorial flagging thresholds in production